Girish

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Website Google Scholar gir-ish girish-b794092a1

Education

Sungkyunkwan University (SKKU), South Korea Sep 2026 – Sep 2030
Integrated M.S.–Ph.D.
Advisor: Prof. Simon S. Woo, DASH Lab

UPES (University of Petroleum and Energy Studies), Dehradun, India Aug 2022 – May 2026
Bachelor of Technology Honours in Computer Science and Engineering (AI-ML)
CGPA: 8.04/10.0

Research Areas: *AI Alignment, Multimodal Learning, Audio LLMs, Deepfake Detection, Speech + Healthcare*
Languages: *English (IELTS: 6.5), Hindi (Native)*

Experiences

Ulster University, Research Assistant — Londonderry, UK *(Remote)*
Advisor: Dr. Muskaan Singh — Dec 2023 – Present

- Working on the SPEECH-D project, funded by Alzheimer's Research UK (ARUK), focused on NLP and speech-based Alzheimer's disease detection using cross-lingual zero-shot learning and cross-accent speech analysis.
- Contributing to a separate research project funded by the Legal Innovation Centre, Ulster University, UK, focused on deep learning and LLM-based solutions for legal-domain applications.

IIIT-Delhi, Undergraduate Research Assistant — New Delhi, India *(Hybrid)*
Advisor: Dr. Arun Balaji Buduru, Orchid Chetia Phukan — June 2024 – May 2026

- Conducting research in deepfake detection, speech emotion recognition, multimodal AI, and Audio Language Models (ALMs), with a focus on developing ML/DL systems for real-world deployment.

Reliance Jio AICoE, ML Engineer — Hyderabad, India *(Remote)*
May 2024 – June 2024

- Evaluated Speech Foundation Models (SFMs) for stress recognition, demonstrating their effectiveness over raw-signal models for physiological signal understanding and cross-domain generalization.

ARTVIEWINGS LLC, AI Research Engineer — San Jose, California *(Remote)*
July 2024 – Aug 2024

- Developed a scalable multilingual audio-visual question-answering system by integrating the MERA framework with ensemble methods, improving performance across eight languages.

Suratec Co., LTD, Computer Vision Intern — Bangkok, Thailand *(Remote)*
June 2024 – July 2024

- Built a video-based golf swing analysis application to detect swing phases and deliver real-time feedback; optimized video processing workflows across multiple input formats.

IBM, AI Intern — *(Remote)*
June 2025 – July 2025

- Developed agentic AI workflows for LLM-powered recommendation systems across multiple domains, integrating reasoning-based agents, domain-specific retrieval, and adaptive recommendation pipelines to improve personalization and decision support.

Publications

Authors marked with an asterisk (*) contributed equally to this work as first authors.
A complete list of publications is available on my Google Scholar profile.

1. **A Paralinguistic Audio–Language Modeling for Zero-Shot Cross-Lingual Alzheimer's Detection**

   Girish*, Mohd Mujtaba Akhtar*, Muskaan Singh, Juliana Gerard, Paula McClean, Kongfatt Wong-Lin

   SPL 2026

2. **Synergizing Zero-Shot Cross-Lingual Alzheimer Detection with Language Invariant Multimodal Bi-Geometric Adversarial Learning**

   Girish*, Mohd Mujtaba Akhtar*, Farhan Sheth, Muskaan Singh, Juliana Gerard, Paula McClean, Kongfatt Wong-Lin

   INTERSPEECH 2026 **Oral**

3. **Towards Detecting Neural Audio Codec Synthesized Heart Sounds**

   Girish*, Orchid Chetia Phukan*, Mohd Mujtaba Akhtar*, Bhavinkumar Vinodbhai Kuwar, Swarup Ranjan Behera, Arun Balaji Buduru

   INTERSPEECH 2026

4. **From Signals to Patterns: Non-Invasive Tuberculosis Detection from Cough Audio using Bandit Weighted Hyperbolic Prototypes**

   Mohd Mujtaba Akhtar*, **Girish***, Sanjam Wadhwa, Muskaan Singh, Ning Ma

   INTERSPEECH 2026 **Oral**

5. **Bridging the Age Gap: Towards Detecting Neural Audio Codec Synthesized Elderly Speech Deepfake**

   Orchid Chetia Phukan*, **Girish***, Mohd Mujtaba Akhtar*, Chi-Chun Lee

   INTERSPEECH 2026 **Oral**

6. **Uc-PrUn: Uncertainty-Calibrated Machine Unlearning using Vision–Language Models for Clinical Decision Support**

   Farhan Sheth, Mohd Mujtaba Akhtar, **Girish**, Muskaan Singh, Juliana Gerard, Paula McClean, Kongfatt Wong-Lin

   ACM Transactions on Computing for Healthcare 2026

7. **Bridging the SEA Gap: An Initial Benchmark for Neural Audio Codec-Synthesized Speech Deepfakes in South-East Asian Languages**

   Orchid Chetia Phukan*, **Girish***, Mohd Mujtaba Akhtar*, Arun Balaji Buduru

   IJCAI 2026

8. **Prosody as Supervision: Bridging the Non-Verbal–Verbal for Multilingual Speech Emotion Recognition**

   Girish*, Mohd Mujtaba Akhtar*, Muskaan Singh

   ACL 2026 Main

9. **Indic-CodecFake meets SATYAM: Towards Detecting Neural Audio Codec Synthesized Speech Deepfakes in Indic Languages**

   Girish*, Mohd Mujtaba Akhtar*, Orchid Chetia Phukan*, Arun Balaji Buduru

   ACL 2026 Findings

10. **HCFD: A Benchmark for Audio Deepfake Detection in Healthcare**

    Mohd Mujtaba Akhtar*, **Girish***, Muskaan Singh

    ACL 2026 Findings

11. **Bridging Attribution and Open-Set Detection using Graph-Augmented Instance Learning in Synthetic Speech**

    Mohd Mujtaba Akhtar*, **Girish***, Farhan Sheth, Muskaan Singh

    EACL 2026 Main

12. **DIVINE: Coordinating Multimodal Disentangled Representations for Oro-Facial Neurological Disorder Assessment**

    Mohd Mujtaba Akhtar*, **Girish***, Muskaan Singh

    EACL 2026 Main **Social Impact Award**

13. **Towards Attribution of Generators and Emotional Manipulation in Cross-Lingual Synthetic Speech using Geometric Learning**

    Girish*, Mohd Mujtaba Akhtar*, Farhan Sheth*, Muskaan Singh

    IJCNLP-AACL 2025 Findings

14. **Curved Worlds, Clear Boundaries: Generalizing Speech Deepfake Detection using Hyperbolic and Spherical Geometry Spaces**

    Farhan Sheth*, **Girish***, Mohd Mujtaba Akhtar*, Muskaan Singh

    IJCNLP-AACL 2025 Main

15. **HYFuse: Aligning Heterogeneous Speech Pre-Trained Representations in Hyperbolic Space for Speech Emotion Recognition**

    Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Swarup Ranjan Behera, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025 **Oral**

16. **Investigating the Reasonable Effectiveness of Speaker Pre-Trained Models and their Synergistic Power for SingMOS Prediction**

    Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Swarup Ranjan Behera, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025

17. **Towards Source Attribution of Singing Voice Deepfake with Multimodal Foundation Models**

    Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Swarup Ranjan, BeheraPriyabrata Mallick, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025

18. **PARROT: Synergizing Mamba and Attention-based SSL Pre-Trained Models via Parallel Branch Hadamard Optimal Transport for Speech Emotion Recognition**

    Orchid Chetia Phukan *, Mohd Mujtaba Akhtar *, **Girish** *, Swarup Ranjan Behera, Sai Kiran Patibandla, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025

19. **SNIFR: Boosting Fine-Grained Child Harmful Content Detection Through Audio-Visual Alignment with Cascaded Cross-Transformer**

    Orchid Chetia Phukan *, Mohd Mujtaba Akhtar *, **Girish** *, Swarup Ranjan Behera, Abu Osama Siddiqui, Sarthak Jain, Priyabrata Mallick, Sai Kiran Patibandla, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025

20. **Towards Machine Unlearning for Paralinguistic Speech Processing**

    Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Shubham Singh, Swarup Ranjan Behera, Vandana Rajan, Muskaan Singh, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025

21. **Towards Fusion of Neural Audio Codec-based Representations with Spectral for Heart Murmur Classification via Bandit-based Cross-Attention Mechanism**

    Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Swarup Ranjan Behera, Priyabrata Mallick, Santanu Roy, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025

22. **SOURCE TRACING OF SYNTHETIC SPEECH SYSTEMS THROUGH PARALINGUISTIC PRE-TRAINED REPRESENTATIONS**

    Girish*, Mohd Mujtaba Akhtar*, Orchid Chetia Phukan*, Drishti Singh, Swarup Ranjan Behera, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

    EUSIPCO 2025

23. **ARE MAMBA-BASED AUDIO FOUNDATION MODELS THE BEST FIT FOR NON-VERBAL EMOTION RECOGNITION?**

    Mohd Mujtaba Akhtar*, Orchid Chetia Phukan*, ***Girish****, Swarup Ranjan Behera, Sanjib, Arun Balaji Buduru, Rajesh Sharma

    EUSIPCO 2025

24. **Are Multimodal Foundation Models All That Is Needed for Emofake Detection?**

    Mohd Mujtaba Akhtar*, ***Girish****, Orchid Chetia Phukan*, Swarup Ranjan Behera, Pailla Balakrishna Reddy, Ananda Chandra Nayak, Sanjib Kumar Nayak, Arun Balaji Buduru

    APSIPA 2025

25. **Rethinking Cross-Corpus Speech Emotion Recognition Benchmarking: Are Paralinguistic Pre-Trained Representations Sufficient?**

    Mohd Mujtaba Akhtar*, Orchid Chetia Phukan*, ***Girish****, Swarup Ranjan Behera, Sanjib, Arun Balaji Buduru, Rajesh Sharma

    APSIPA 2025

26. **Beyond Speech and More: Investigating the Emergent Ability of Speech Pre-Trained Models for Classifying Physiological Time-Series Signals**

    Orchid Chetia Phukan*, Swarup Ranjan Behera*, ***Girish****, Mohd Mujtaba Akhtar, Sanjib, Arun Balaji Buduru, Rajesh Sharma

    APSIPA 2025

27. **Investigating Polyglot Speech Foundation Models for Learning Collective Emotion from Crowds**

    Orchid Chetia Phukan*, ***Girish****, Mohd Mujtaba Akhtar*, Panchal Nayak, Priyabrata Mallick, Swarup Ranjan Behera, Parabattina Bhagath, Pailla Balakrishna Reddy, Arun Balaji Buduru

    APSIPA 2025

28. **Strong Alone, Stronger Together: Synergizing Modality-Binding Foundation Models with Optimal Transport for Non- Verbal Emotion Recognition**

    Orchid Chetia Phukan, Mohd Mujtaba Akhtar*, ***Girish****, Swarup Ranjan Behera, Sishir Kalita, Arun Balaji Buduru, Rajesh Sharma, SR Mahadeva Prasanna

    ICASSP 2025

29. **NeuRO: An Application for Code-Switched Autism Detection in Children**

    Mohd Mujtaba Akhtar*, ***Girish****, Orchid Chetia Phukan*, Muskaan Singh*

    INTERSPEECH Show & Tell 2024

Research Software & Libraries

- **Neural Codecs:** — Library for generating neural codec-processed audio datasets using multiple codec models for speech synthesis, deepfake detection, and audio representation research.
- **SFM Models** and **MFM Models** — Toolkits for extracting embeddings from speech, music, and general audio signals using large-scale pre-trained speech, music, and multimodal foundation models.

Technical Skills

Frameworks and Libraries: TensorFlow, Keras, PyTorch, OpenCV, LangChain

Programming Languages: Python, C, Java, SQL

Web Technologies: HTML, JavaScript, FastAPI, Flask

Technologies: Git, Linux, AWS, Docker

Selected Projects & Innovation

Memoir: Adaptive Audiobooks for Memory Care *Selected Project, Idiap Create Challenge 2026 — Idiap Research Institute, Switzerland* *August 2026*

- Selected for the nine-day Idiap Create Challenge 2026 at the Idiap Research Institute, Martigny, Switzerland, to develop *Memoir*, an AI-powered adaptive audiobook platform for personalized memory-care support.

Selected Projects

Helix: Versatile AI Assistant GitHub

- AI-Powered Smart Responses: Uses OpenAI GPT, LangChain, and Pinecone for real-time, context-aware interactions, improving customer engagement and automation.
- Versatile & Scalable: Easily deploys across industries like healthcare, retail, and finance with minimal code changes, ensuring seamless adaptability.
- Omnichannel & Fast: Connects via WhatsApp (Twilio) and phone calls, with a FastAPI backend and async MongoDB for quick, efficient responses.
- **Tools Used: LangChain, FastAPI, MongoDB, Twilio.**

Multimodal Personality Prediction Using Contrastive Learning GitHub

- Built a neural network with two processing paths that uses contrastive learning to predict personality traits (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) from audio and video data.
- Optimized the system for better integration of multimodal data, ensuring accurate and reliable personality predictions from real-world audio and video inputs.
- **Tools Used: TensorFlow, Hugging Face Transformers.**

Golf Phase Detection and Analysis Application. GitHub

- Developed algorithms to detect and classify golf swing phases (setup, backswing, downswing, ball impact, and follow-through) with high accuracy.
- Designed a user-friendly interface for players and coaches, enabling easy review and real-time analysis across different video formats.
- **Tools Used: OpenCV, MediaPipe, FFmpeg, Streamlit**

Certificates and Achievements

- **Awarded the Social Impact Award at EACL 2026 for the paper** *"DIVINE: Coordinating Multimodal Disentangled Representations for Oro-Facial Neurological Disorder Assessment."*
- Awarded the *Silver Medal* and secured *First Runner-Up* position in the Technical Idea Presentation at ICE Day, UPES, Dehradun (22 October 2024), for a concept related to speaker verification.

Research Grants and Funding

- Secured over **£19,996** in research funding as the **lead applicant** for two projects at Ulster University: the *SPEECH-D* project, funded by Alzheimer's Research UK (ARUK), and a legal-domain AI research project funded by the Legal Innovation Centre, Ulster University.
- Awarded a **€ 1,000** ISCA travel grant to attend Interspeech 2025 in Rotterdam, The Netherlands.

Academic Service

- **ISCA Team Volunteer:** Managing script development and co-hosting episodes on speech technology and AI advancements.
- **Reviewer:** ICASSP 2026, ACM Multimedia 2026.
- **Program Committee Member:** EMNLP 2026.

Conference Attended

- **INTERSPEECH 2025**, Rotterdam, Netherland